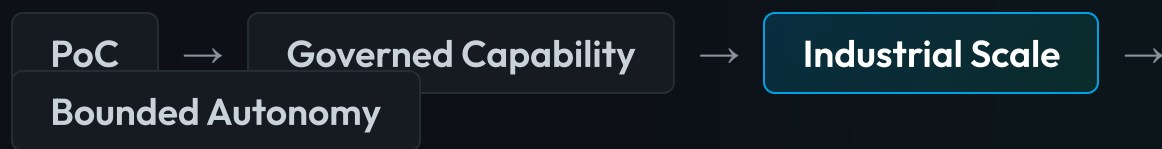


Current State of AI

Solution Architecture Point of View

From AI experiments to governed, measurable, context-rich AI



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AI crossed the usefulness threshold. Enterprises have not crossed the control threshold.

What works today	What breaks at scale	Architecture response
GenAI, coding agents, MCP tools, copilots, skills, PoCs	Governance, traceability, determinism, integration, cost, ownership	Control plane, cards, semantic layer, evaluation, AI FinOps

- ▶ Tooling is ahead of the operating model
- ▶ Customers merge GenAI, agentic use cases, MCP, skills, platforms — without lifecycle governance
- ▶ The missing layer is not another framework — it is **lifecycle governance** and **measurable execution**
- ▶ Governance must be baked into agents, not attached after rollout

Six workstreams for industrial AI

1 AGENTIC PLATFORM & GOVERNANCE

Control plane, cards, lifecycle, policy, evaluation, observability, HITL, cost controls

2 SKILLS & CAPABILITY GOVERNANCE

Skill inventory, evaluation, ownership, reuse tracking, governed enterprise capabilities

3 BROWNFIELD & MODERNIZATION

Knowledge recovery, spec-driven execution, traceability, evidence-backed transformation

4 SEMANTIC LAYER & EVIDENCE GRAPH

Governed context substrate, traceability, precise retrieval, cost reduction through specificity

5 AI FINOPS & MODEL PORTFOLIO

Cost visibility, model tiering, spend limits, cost-per-outcome economics

6 ADOPTION BEYOND ENGINEERING

Manager co-working, role-specific playbooks, delegation, review, governance literacy

Most agentic rollouts fail structurally, not technically

ROLE-EMULATING AGENTS

Digital copies of human roles instead of redesigned delegation patterns

GOVERNANCE AS AFTERTHOUGHT

Acceptable in PoCs, fatal at scale

DATA LAYER IN PARALLEL

Data lakes are not enough for agentic context

PREMATURE PLATFORM

Expensive target architectures before real use cases teach requirements

PREMATURE TECHNOLOGY

Framework debates before strategy, IP, and governance questions

MCP standardized interoperability, not governance.

The target architecture: control plane, runtime governance, semantic layer

Experience

Chat, IDE, portal, voice, email, workflow, API

Meet users where they work

Control Plane

Cards, lifecycle, policy, HITL, evaluation, telemetry, cost, identity

Make governance executable

Agent Runtime

Skills, MCP tools, coding agents, multi-agent flows

Run capabilities safely

Semantic Layer

Business concepts, code facts, specs, tickets, SME feedback, traces

Governed context & traceability

Systems of Record

Git, Jira, CI/CD, CRM, ERP, legacy systems, data platforms

Connect AI to enterprise reality

The market over-focuses on the runtime. Architectural leverage sits in the **control plane** and **semantic layer**.

Agentic Platform & Governance

Workstream 1

Primitive	Purpose
Skill Card	What a skill does, reads/writes, owner, evaluation
Tool Card	Governance object for MCP tools and integrations
Agent Card	Purpose, boundaries, authority, evaluations, KPIs, policies
Control Plane	Lifecycle, evaluation, policy, observability, HITL, cost
Semantic Layer	Runtime context, evidence, feedback, traces, business rules

BUILD-OUT SEQUENCE

1. Skills under observation
2. MCP legitimized within BUs
3. Agents as BU-crossing coordination
4. Standing decisions / goal-oriented agents

Governance is not a committee. In industrial agentic systems, **governance is a runtime property.**

Agentic Suitability & TCO

Workstream 1

Low decision effort

High decision effort

High volume

Rule-based automation, scripts, RPA

Strong candidate for agentic delegation

Low volume

Do not automate

Only if risk or value is high

- ▶ High volume + low complexity → deterministic automation
- ▶ Agentic value strongest where **frequency meets judgment**
- ▶ Autonomy increases TCO; validate suitability against cost & governance

Don't over-agentify. If the task is deterministic, use deterministic automation.

Skills & Capability Governance

Workstream 2

SKILLS ARE STICKY NOTES IN THE WILD

- ▶ Teams and individuals create skills locally — fast, useful, **ungoverned**
- ▶ No inventory: nobody knows how many skills exist, what they do, or who owns them
- ▶ No evaluation: skills are used but **never measured for quality, cost, or drift**
- ▶ No reuse tracking: the same capability is reinvented across teams
- ▶ Skills become **shadow IT capabilities** — powerful but invisible to the organization

Without evaluation, there is no long-term benefit. Skills that are never measured cannot improve, compound, or be trusted at scale.

THE GOVERNANCE GAP

Invisible proliferation — skills multiply without central visibility, creating ungoverned enterprise IT capabilities

No quality signal — without evaluation, bad skills survive as long as good ones

No ownership — when the creator leaves, the skill becomes orphaned but still runs

No cost attribution — token spend is invisible and unmanaged per skill

From unmanaged skill usage → governed skills library

Brownfield & Modernization

Workstream 3

*The problem is not the target technology. The problem is **recovering business logic** from legacy systems and preserving it through transformation.*

CURRENT REALITY

- ▶ Mixed results across teams
- ▶ No repeatable approach yet
- ▶ Many successes are senior experts doing bespoke work
- ▶ **Heroic seniors are not a factory**

PLAYBOOK RESPONSE

ANALYZE

Knowledge base, dependency map, risk classification, behavioral baseline

PLAN

Target architecture, migration strategy, execution roadmap

EXECUTE

Specs, generated code, tests, validation evidence,
review feedback

Spec-driven execution loop

Workstream 3

Source Evidence

Specification

Generated Artifact

Automated Validation

Deploy / Archive

Feedback into KB

Human Review Gate

Human Review Gate

Spec Review Loop

TRACEABILITY CHAIN

Source → KB → Spec → Code → Test → Deploy

THE MISSING FEEDBACK LOOP

Most projects have no systematic process to **create, review, and improve specs**. Specifications are written once and never revisited. This is where the senior hero pattern comes from — only experienced architects intuitively close this loop.

Every generated artifact must carry receipts: what evidence was consumed, what model produced it, what gate accepted it, and what feedback was captured.

Semantic Layer & Evidence Graph Workstream 4

*The semantic layer is not a data lake, not GraphRAG, and not a BI semantic model. It is the **governed context substrate for AI**.*

EXAMPLE: AN SDLC EVIDENCE GRAPH CAPTURES

Code facts & business rules

Requirements & specifications

Tickets, SOPs & operational evidence

SME feedback & human corrections

Agent traces & tool outputs

Validation & review decisions

This is one domain graph. The same pattern applies to sales, operations, compliance, and any governed AI workflow.

WHY BUILD THE SEMANTIC LAYER

Quality: Precise context produces better artifacts. Less hallucination, more grounded outputs, easier to validate.

Traceability: Every AI output links back to evidence. Auditable, explainable, governable.

Cost reduction: Smaller, specific prompts mean fewer tokens. Route to cheaper models when context is strong. Cache reusable context.

Compounding value: Every correction, review, and feedback enriches the graph. AI gets better over time, not just faster.

AI FinOps & Model Portfolio

Workstream 5

Cost context	Meaning
~\$40– 50/dev/day	Workshop mode: heavy refactoring, no optimization
~6avg/12 p90	Industry benchmark expectation (comparison, not target)

- ▶ Being more efficient is **not automatically bottom-line value**
- ▶ The stronger case: retire technical debt for **bounded cost**
- ▶ “Kill a decade-old backlog for \$250k” is a business decision

Encoders Classification, retrieval, routing

Small Templated generation, formatting

Mid-tier Complex generation, analysis

Frontier High-uncertainty reasoning

Frontier models should be **scarce resources**, not default infrastructure.

Adoption Beyond Engineering

Workstream 6

CLAUDE COWORK FOR MANAGERS

- ▶ Workshop for non-technical users: very well received
- ▶ High-leverage adoption path
- ▶ AI cannot remain developer-only
- ▶ Managers need to **delegate, review, govern, and supervise** AI-supported work

Role group	Focus
Managers	Delegation, review, supervision, KPI interpretation
Business Analysts	Better requirements, spec quality, business-IT alignment
Architects	Traceability, target architecture, governance, modernization
QA	AI-assisted test generation, validation, review gates

Role group

Focus

Operations

Incident support, remediation,
workflow automation

Developer-only adoption



Organizational AI co-working

Roadmap: visibility → repeatability → scale → autonomy

Workstream	6 months	1 year
1. Agentic Platform	Inventory skills, tools, MCPs, agents. Introduce Cards. Measure governed vs ungoverned.	BU-level MCP registries evaluation observability HITL, ownership cost tracking.
2. Skills & Capabilities	Baseline work profiles. Seed skills library from hackathon artifacts.	Reusable evaluated skills, ownership regression checks, coding-agent governance
3. Brownfield & Modernization	Select 1–2 pilots. Build	Run source → KB →

first KB from
code, DBs,
tickets,
SOPs, SME
validation.

spec →
artifact →
validation
review loop
Prove
repeatable

4. Semantic Layer

Define
evidence
model.
Connect
initial tool
graphs.
Capture
feedback &
traces.

Use for
retrieval,
prompt
grounding
spec
generative
validation
traceability
Introduce
encoders

5. AI FinOps

Make cost visible.
Track cost per work profile.

Spend limit model tiering, prompt optimization, caching, cost per artifact.

6. Adoption

Productize Cowork for managers.
Teach delegation, review, governance literacy.

Role-specific playbook for BAs, architect QA, production operation

Three leadership decisions

FUND GOVERNANCE PRIMITIVES

Cards, registries, control plane MVP, evaluation, observability, HITL, cost telemetry

FUND SEMANTIC LAYER & MODERNIZATION

Evidence graph, spec-driven loop, feedback capture, validation, traceability

FUND ROLE-BASED ADOPTION

AI-SDLC baselines, manager coworking, role playbooks, cost-per-outcome dashboards

Visibility



Governed Repeatability



Industrial Scale



Bounded Autonomy

The winners will not be the companies with the most AI access.
They will be the companies with the best governed AI operating model.

What we actually see in the market

TECHNOLOGY PUSH, NOT BUSINESS PULL

Most companies are head over heels scrambling to build AI platforms. We see a common lack of business pull — teams build because they can, not because a business outcome demands it.

AI COST IS SEVERELY UNDERAPPRECIATED

Workshop costs feel invisible today. Once adoption scales beyond engineering, unrestricted frontier-model access will hit budget walls nobody is planning for.

NO SEMANTIC LAYER UNDERSTANDING

Very few organizations recognize the semantic layer as the strategic asset it is. Most conflate it with data lakes or RAG — missing the governance, traceability, and cost benefits.

THIS WILL CHANGE FAST

OpenAI Frontier and Claude Cowork / Claude Code are making agentic AI accessible to non-engineers. When managers and business users adopt AI co-working, the demand for governance, cost control, and semantic context will become urgent — not optional.

The question is not whether your organization will need governed AI. It is whether you will be ready when adoption outpaces your operating model.

Thank you!



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The cat has spoken. Now it's your turn.

Speaker notes